

boundaries of distributed computing solutions. These were the early days of cloud computing.

As per Brewer, the bounding principles for any distributed service are consistency, availability and partition tolerance (CAP). Any architecture can comply with two of these three principles but not with all three at the same time.

Open data platforms

When open source first became a buzzword in the IT world, it was a revolutionary idea that reduced IT costs and changed the development ecosystem into a highly collaborative one. The current game-changing thought is 'open data'. The idea is to develop an open data marketplace without compromising security and data privacy.

Hortonworks, IBM and Pivotal have been working on the Open Data Platform (ODP) for a few years and have released their own ODP architecture like ODPi and IBM Open Platform with Apache Hadoop (IOP), which are available on IBM Power Systems and Pivotal HD.

One important advantage of the ODP is that it offers new solutions for Big Data architecture with better time to market results. Today, e-governance, the financial industry, retail marketing and many other industries use open data platforms for targeted marketing and trend analysis.

Data lineage

It is important to trace the complete life cycle of data to understand the data source, how it travelled in different feed formats, how it got modified and who owns data in its various forms.

The data life cycle is termed data lineage and is useful for data debugging and understanding various metrics related to data architecture. Traditionally, to understand data duplicacy, errors in data and data governance, data provenance was used as a tool for tracking data.

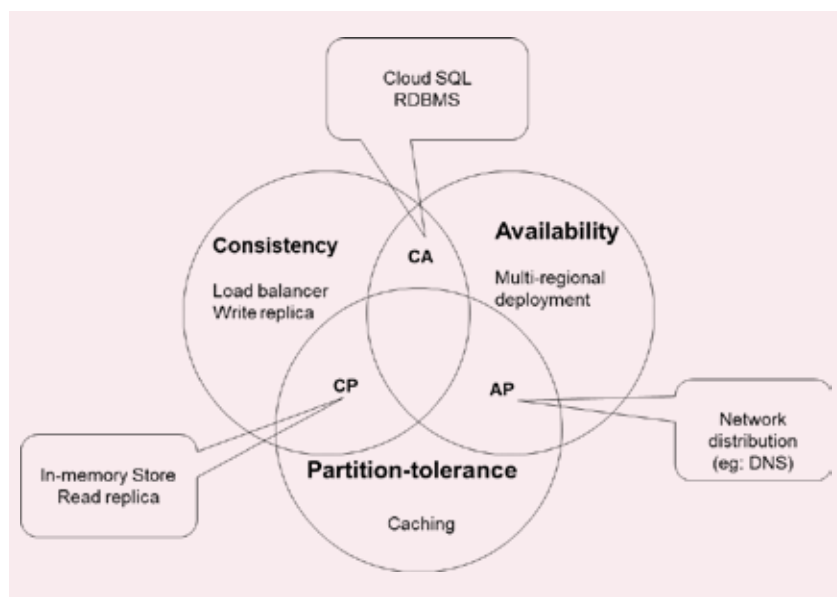


Figure 2: CAP theorem for data solutions

Data lineage helps in decision making for data analytics activities, data analysis and debugging in data architecture. It is managed through metadata to store and retrieve data origin and transformation details. Data lineage tools like Collibra, Talend Data Fabric, Apache Nifi and Apache Atlas show data in its transformation and origin through metadata extraction, as well as the data flow of resilient distributed datasets (RDDs) using graphs and other visualisation maps.

Five W's are important for metadata handling — what is the meaning of the data, who is using the data, when was it captured (first and subsequent changes), when is it being used (first and recent use), and where is the data stored/transformed. This helps to debug data errors or data duplication (de-duplication) and data validation during Big Data processing.

DataOps for a data lake solution in the AWS platform

Building a resilient data platform in the cloud is a tedious task as it must fit into the DevOps architecture to automate infrastructure provisioning,

data export, platform setup, data movement and processing, and data validation and testing in order to reduce the repetitive manual effort of data engineering.

A data lake combines a data warehouse and database setup in AWS, developing a DataOps solution to accelerate data processing activities. DataOps aims at developing a customised DevOps and AIOps solution for data platforms, which combines the activities of data management, data migration, data preparation (pre-migration), data validation (post migration), data analytics, platform setup and data governance.

Autonomous database: The next-generation cloud database

A DBA or database administrator handles database tuning, security activities, backups, disaster recovery, server/platform updates, health checks and all the other management and monitoring activities of databases. When a cloud platform is used for application and database development, these activities become critical for improving security, performance and cost efficiency. Oracle has developed a